College Timetable Generator

What the app does, and what has been built so far · 21 October 2026

The app makes a weekly class timetable for a college in which no teacher, class or room is ever booked twice at the same time. A timetable coordinator enters the college's details and presses Generate. Everything is saved in their web browser, so there is no login.

1. Features

The app works in four steps. Each has a one-line hint, and tricky fields have an (i) tip.

Step 1 Set up the week

- **Working days.** Pick any days from Monday to Sunday.
- **Periods.** Set each period's name, start time, end time and whether it is a teaching period or a break. **Generate periods** fills the list from a start time, a period length and a count. Every day uses the same periods.
- **Lunch.** Mark the periods where lunch may fall. Every section and every teacher gets at least one of them free each day.
- **Mistake checks.** A period with no name, a missing time, an end before its start, or a start before the previous period ends is highlighted. The step also needs at least one day, one teaching period and one lunch period.

Changing the week clears the current timetable, because the old one no longer fits the new days and periods.

Step 2 Add the college's data

- **Sample college.** One button loads a complete example college to try the app with.
- **Excel template.** Download a workbook with one sheet each for departments, teachers, rooms, sections and courses. Fill it in and upload it. Separate CSV files, one per sheet, also work.
- **Clear import errors.** If anything is wrong, nothing is imported and each problem is listed by row, e.g.: *Courses row 7: teacher "Sharma" not found*. A correct file asks before replacing the current data.
- **Export.** Download the current data as an Excel file in the same layout.
- **Edit tables.** Add, change or delete courses, sections, teachers, rooms and departments on screen.
- **Rooms.** Each room is a classroom or a lab. A room that belongs to a department is used only by that department; a room with no department is shared by all. A lab course can ask for one particular lab.
- **Lab batches.** A section can be split into batches for labs. Each batch gets its own lab classes, different batches can be in different labs at the same time, and each batch can have its own lab teacher.
- **Data checks.** Serious problems block the next step: a missing teacher or room, a class with no suitable room, or a lab longer than any run of back-to-back periods. A section or teacher with more classes than free periods gets a warning.

Step 3 Make the timetable

- **Choice of solver.** *Auto* (recommended) uses the exact solver when the server is running and the fast one otherwise. *Exact* finds the best possible timetable but needs the server. *Fast* runs in the browser, works offline, and gives a good timetable in a few seconds.

- **Time limit.** A slider from 5 to 60 seconds (20 by default). A coloured dot shows whether the server is running, and a timer counts the seconds while the timetable is being made.
- **Automatic switch.** In Auto mode, if the server is off, too slow, fails, or finds that the data cannot fit, the fast solver takes over. When the data cannot fit, it still places as many classes as it can.
- **Classes that didn't fit.** Each one is listed with the reason, such as the teacher or room that was already busy.
- **Checked result.** Every timetable is checked for clashes before it is shown, whichever solver made it, and a badge shows the result, e.g. *Verified: 0 clashes*.

Step 4 View, adjust and export

- **Weekly grid.** Switch between the timetable of a section, a teacher or a room. Search for one by name or step through them with previous and next. Labs stretch across their periods, each subject has its own colour, and labs of different batches at the same time sit in the same cell.
- **Class details.** Hover over a class for a short summary; click it to see subject, section, batch, teacher, room and time.
- **Lock and regenerate.** Lock a class to keep it in its slot (it gets a lock icon and a thick border). **Regenerate** remakes everything else around the locked classes. Locks that no longer make sense, for example because a room was deleted, are removed with a warning. **Clear all locks** removes every lock.
- **Statistics.** The quality score split by rule; how full each room is, with a total for shared rooms; and each teacher's classes per week and per day, sortable, with the busiest and least busy marked.
- **Printing.** Print the timetable on screen, or every section's timetable on its own landscape A4 page. Subject colours stay, buttons and menus are left out, and a heading line such as the college name and semester goes at the top. Choose *Save as PDF* in the print window to get a file.
- **Spreadsheet download.** One row per class with section, batch, day, periods, time, subject, type, teacher and room. It opens directly in Excel.

2. Rules every timetable follows

Never broken. If a class can't be placed without breaking one of these, it is listed as not placed instead.

- A teacher teaches one class at a time.
- A section has one class at a time. A whole-section class clashes with all of its batches; two batches clash only with themselves.
- A room holds one class at a time.
- A class goes in the right kind of room, in a room its department may use, and in the particular lab a course asks for.
- Every class is placed exactly once.
- A lab fills back-to-back teaching periods on one day and never runs through a break.
- Nothing is placed in a break.
- Every section and every teacher gets a free lunch period every day.

Kept to where possible. Each of these adds penalty points to the score; a lower score is a better timetable.

What is avoided	Points
The same subject taught more than once in a day	10 for each extra lecture
A teacher teaching more than 3 periods in a row	5 for each period past 3
Empty periods in the middle of a section's day (lunch periods don't count)	2 for each empty period
One batch sitting idle while another batch of the same section is in a lab	3 for each idle period

3. Known limits

- The fast solver always uses its full time limit unless it finds a perfect timetable, and two runs can give slightly different results.
- The exact solver needs about 2 seconds before it has any answer for the sample college, so a very short time limit can fail. In Auto mode the fast solver then takes over.
- In Exact mode with the server off, the app shows an error and keeps the previous timetable. Only Auto switches solvers by itself.
- If a lock leaves no room for other classes, those classes show up as not placed; the lock itself is kept.
- Printed timetables don't show which classes are locked.
- Running the app needs two programs started on the computer: the website and the solver server. Without the server, only the fast solver is available.

4. Progress

12 of 15 steps are done, so the app works from start to end. Green is done, grey is still to do.

1. **Project setup** Website, solver server and automatic tests.
2. **College data format** How a college is described, plus the sample college.
3. **Clash checker and score** Finds every clash and counts smaller problems.
4. **Fast solver** Makes a timetable in the browser.
5. **Exact solver** Finds the best timetable on the server.
6. **Guided steps** Four-step layout and the week setup screen.
7. **Data entry** Excel import and export, edit tables, data checks.
8. **Generate screen** Solver choice, automatic switch, checked results.
9. **Timetable view** Grid for any section, teacher or room.
10. **Statistics** Quality score, teacher workload, room use.
11. **Printing and download** Print to PDF and spreadsheet download.
12. **Lock and regenerate** Keep chosen classes, remake the rest.
13. **Polish** Fixes after a trial with a real user.
14. **Drag and drop** Move classes by dragging them in the grid.
15. **Put online** Run the app without installing anything.